

```

# Part A
# Dictionaries

# 1. Create an empty dictionary
inventory = {}

# 2. Store the first product details in variables
ProductNo1_name = "Portable Projector"
ProductNo1_quantity = 4
ProductNo1_price = 18500
ProductNo1_releaseYear = 2021

# 3. Add the first product details to inventory
inventory["ProductNo1_name"] = ProductNo1_name
inventory["ProductNo1_quantity"] = ProductNo1_quantity
inventory["ProductNo1_price"] = ProductNo1_price
inventory["ProductNo1_releaseYear"] = ProductNo1_releaseYear

# 4. Store the second product details in variables
ProductNo2_name = "Electric Kettle"
ProductNo2_quantity = 7
ProductNo2_price = 3200
ProductNo2_releaseYear = 2024

# 5. Add the second product details to inventory
inventory["ProductNo2_name"] = ProductNo2_name
inventory["ProductNo2_quantity"] = ProductNo2_quantity
inventory["ProductNo2_price"] = ProductNo2_price
inventory["ProductNo2_releaseYear"] = ProductNo2_releaseYear

# 6. Display the products present in inventory

print("Products present in inventory:")

for product, details in inventory.items():
    print(product, ":", details)

Products present in inventory:
ProductNo1_name : Portable Projector
ProductNo1_quantity : 4
ProductNo1_price : 18500
ProductNo1_releaseYear : 2021
ProductNo2_name : Electric Kettle
ProductNo2_quantity : 7
ProductNo2_price : 3200
ProductNo2_releaseYear : 2024

# 7. Check the release years of both products

print("\nChecking release years:")

```

```
print("ProductNo1_releaseYear in inventory:",  
      "ProductNo1_releaseYear" in inventory)  
  
print("ProductNo2_releaseYear in inventory:",  
      "ProductNo2_releaseYear" in inventory)
```

```
Checking release years:  
ProductNo1_releaseYear in inventory: True  
ProductNo2_releaseYear in inventory: True
```

8. Delete release year of both products

```
del inventory["ProductNo1_releaseYear"]  
del inventory["ProductNo2_releaseYear"]  
  
print("\nInventory after deleting release years:")  
print(inventory)
```

```
Inventory after deleting release years:  
{'ProductNo1_name': 'Portable Projector', 'ProductNo1_quantity': 4,  
'ProductNo1_price': 18500, 'ProductNo2_name': 'Electric Kettle',  
'ProductNo2_quantity': 7, 'ProductNo2_price': 3200}
```

Part B
Tuples

1. Create a tuple containing product prices

```
prices = (180, 450, 275, 600, 125, 350, 220)
```

```
print("Prices:", prices)
```

```
Prices: (180, 450, 275, 600, 125, 350, 220)
```

2. Find highest and lowest price

```
highest_price = max(prices)  
lowest_price = min(prices)  
  
print("Highest price:", highest_price)  
print("Lowest price:", lowest_price)
```

```
Highest price: 600  
Lowest price: 125
```

3. Calculate total sum of all prices

```
total = sum(prices)
```

```
print("Total sum of all prices:", total)
```

Total sum of all prices: 2200

4. Convert tuple into sorted list in ascending order

```
sorted_prices = sorted(prices)
```

```
print("Sorted list:", sorted_prices)
```

Sorted list: [125, 180, 220, 275, 350, 450, 600]

5. Trying to modify an element in the tuple

```
try:
```

```
    prices[0] = 500
```

```
except TypeError:
```

```
    print("Tuple cannot be modified because tuples are immutable.")
```

Tuple cannot be modified because tuples are immutable.